Katugampola Fractional Derivative

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March 4, 2026

1 Introduction

The Katugampola fractional derivative is Equation 1:

$$D_a^\alpha f(x) = \frac{1}{\Gamma(m-\alpha)} \frac{d^m}{dx^m} \int_0^x (x-t)^{m-\alpha-1} f(t) dt, \quad \text{where } m = \lceil \alpha \rceil \text{ and } \alpha \in (0,1) \tag{1}$$

2 Derivation of Katugampola Fractional Derivative

To derive Equation 1, we start with Equation 2:

$$I_{0+}^\alpha f(x) = \frac{1}{\Gamma(\alpha)} \int_0^x (x-t)^{\alpha-1} f(t) dt \tag{2}$$

where $0 < \alpha < 1$. Then, we have:

$$D_a^\alpha f(x) = \frac{1}{\Gamma(m-\alpha)} \frac{d^m}{dx^m} I_{0+}^\alpha f(x) = \frac{1}{\Gamma(m-\alpha)} \frac{d^m}{dx^m} \left(\frac{1}{\Gamma(\alpha)} \int_0^x (x-t)^{\alpha-1} f(t) dt \right) \tag{3}$$

This leads to Equation 1.

3 Properties

3.1 Linearity

The derivative satisfies :

$$D_a^\alpha(cf + g) = cD_a^\alpha f + D_a^\alpha g \tag{4}$$

where c is .

4 Conclusion

The Katugampola fractional derivative

Equations (1), (4), and (??) illustrate key aspects of the Katugampola fractional derivative.